

Hoigi Seo

Homepage: seohoiki3215.github.io

LinkedIn: www.linkedin.com/in/hoigiseo

Google Scholar: <https://scholar.google.com/citations?user=J5w82IEAAAJ&hl>

Email: seohoiki3215@snu.ac.kr

Mobile: +82-10-3222-3295

EDUCATION

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- **Seoul National University** Seoul, Republic of Korea
B.E. in Electrical and Computer Engineering March 2018 - February 2024
 - **Seoul National University** Seoul, Republic of Korea
Ph.D. in Electrical and Computer Engineering March 2024 - Present

RESEARCH INTEREST

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- **Multi-Modal Generative Models:** Multi-modal large language model, Text-to-image / video diffusion model
 - **Efficient Model Training & Inference:** Quantized model training, Diffusion model acceleration
 - **Efficient Agentic System:** Small Language Models (SLMs), Multi-Agent System, Agentic Swarm

PUBLICATIONS (SELECTED)

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- **Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules**
J Bang*, D Moon*, **Hoigi Seo**, S Hong & S Y Chun. (*co-first authors)
International Conference on Machine Learning (ICML), 2026.
 - **Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models**
Hoigi Seo*, B H Lee*, J Cho, S Lim & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
 - **Training free, Perceptually Consistent Low-Resolution Previews with High-Resolution Image for Efficient Workflows of Diffusion Models**
W Jeong*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
 - **Training-free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers**
W Jeong*, K Lee*, **Hoigi Seo** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR) Highlight, 2026.
 - **On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models**
Hoigi Seo*, D U Kang*, H Cho, J Lee & S Y Chun. (*co-first authors)
Neural Information Processing Systems (NeurIPS), 2025.
 - **Skrr: Skip and Re-use Text Encoder Layers for Memory Efficient Text-to-Image Generation**
Hoigi Seo*, W Jeong*, J Seo & S Y Chun. (*co-first authors)
International Conference on Machine Learning (ICML), 2025.
 - **Efficient Personalization of Quantized Diffusion Model without Backpropagation**
Hoigi Seo*, W Jeong*, K Lee & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
 - **INTRA: Interaction Relationship-Aware Weakly Supervised Affordance Grounding**
J H Jang*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
European Conference on Computer Vision (ECCV), 2024.
 - **BeyondScene: Higher-Resolution Human-Centric Scene Generation with Pretrained Diffusion**
G Kim*, H Kim*, **Hoigi Seo***, D U Kang* & S Y Chun. (*co-first authors)
European Conference on Computer Vision (ECCV), 2024.

PREPRINTS

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- **Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs**
Hoigi Seo, B H Lee, M Kim, J Lee & S Y Chun.
NeurIPS Under review, 2026.
 - **Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis**
Hoigi Seo, H Bae, B H Lee, J Cho, J Lee, Y Kim, S Y Jeon, J H Jang, H Kim & S Y Chun.
NeurIPS Under review, 2026.
 - **HYPERION: Joint-Hierarchy Hyperbolic Fine-Tuning for Histopathology Slide Representations**
J H Jang, H Kim, S Y Jeon, Y Kim, J Cho, J Lee, **Hoigi Seo**, H Bae, B H Lee & S Y Chun.
NeurIPS Under review, 2026.
 - **Recovering Diversity in Distilled Diffusion Models via Low-Rank Policy-Optimized Merging**
Hoigi Seo, W Han, J Lee & S Y Chun.
WACV Under review, 2027.

- **Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation**
Hoigi Seo*, J Bang*, H Lee*, J Lee, B H Lee & S Y Chun. (*co-first authors)
arXiv preprint, 2025.
- **DITTO-NeRF: Diffusion-based Iterative Text To Omni-directional 3D Model**
Hoigi Seo*, H Kim*, G Kim* & S Y Chun. (*co-first authors)
arXiv preprint, 2023.

SKILLS SUMMARY

- **Programming Languages:** Python, C, C++, Bash
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **Distributed Training:** PyTorch DDP / FSDP, DeepSpeed, HuggingFace Accelerate
- **Inference & Serving:** vLLM, quantization (GPTQ / AWQ / bitsandbytes)
- **MLOps & Tooling:** Docker, Git, Linux, Weights & Biases, NVIDIA GPU cluster (A100 / H100)
- **Languages:** English (professional), Korean (native)

PATENTS

- **Speculative Encoding-Based Method and Apparatus for Whole Slide Image Analysis.**
S Y Chun, **Hoigi Seo**, H Bae, B H Lee, J Cho, & J Lee
Korean Patent, Filed. No. 10-2026-0107545, 2026.
- **Electronic Apparatus Including an Artificial Intelligence Model for Generating an Image and Operating Methods.**
S Y Chun, W Jeong, K Lee & **Hoigi Seo**
Korean Patent, Filed. No. 10-2026-0103476, 2026.
- **Electronic Apparatus for Generating Images and Operating Methods.**
S Y Chun, W Jeong & **Hoigi Seo**
Korean Patent, Filed. No. 10-2026-0106226, 2026.
- **Device and Method for Affordance Grounding.**
S Y Chun, J H Jang & **Hoigi Seo**.
Korean Patent, Filed. No. 10-2025-0029227, 2025.
- **Device and Method for Extracting High-Dimensional Gene Features.**
S Y Chun, **Hoigi Seo**, H Bae, D U Kang & H Kim.
Korean Patent, Filed. No. 10-2025-0014990, 2025.
- **Higher-Resolution Human-Centric Scene Generation With Pretrained Diffusion.**
S Y Chun, G Kim, H Kim, **Hoigi Seo** & D U Kang.
Korean Patent, Filed, No. 10-2024-0115484, 2024 / U.S. Patent Application, Filed, No. 19/019,060, 2025.

HONORS AND AWARDS

- Compute Transparency Champion Award (CVPR26) | June 2026
- Excellent Research Talent Fellowship (Korea Research Foundation) | September 2025
- Brain Korea 21 Scholarship (Korea Research Foundation) | March 2024 - Present
- Academic Excellence Scholarship (Seoul National University, ECE) | September 2023

RESEARCH PROJECT EXPERIENCE

- **Uncertainty-aware World Model and VLA Model** Project Lead
Korea Electronics Technology Institute (KETI), Republic of Korea July 2026 - Present
 - Building uncertainty-aware world model and VLA model that can estimate current state’s uncertainty and account for the uncertainty for the next action and prediction.
- **Megabase-scale DNA foundation model** Project Lead
NVIDIA Academic Grant (32K A100 hours) April 2026 - Present
 - Building foundation models for Megabase-scale DNA foundation model for structural variant detection.
- **ecDNA Detection** Project Lead
Ministry of Science and ICT, Republic of Korea September 2023 - Present
 - Extra-chromosomal DNA (ecDNA) detection and associated gene discovery from FISH images and gene profiles.
- **AI for Human-Machine Teaming** Project Participant
United States Air Force & IITP, Republic of Korea March 2025 - Present
 - Developing human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.
- **Efficient and Performative Pathology Foundation Model** Project Participant
National IT Industry Promotion Agency (NIPA), Republic of Korea January 2026 - June 2026
 - Building efficient digital pathology inference method and train 7B-parameter histopathology multi-modal large language model.
- **Efficient Personalization** Project Participant
Samsung Mobile Experience (MX) March 2024 - May 2025
 - Memory and time efficient text-to-image synthesis and personalization for on-device deployment.
- **Industrial Defect Segmentation** Project Participant
Samsung Device Solution (DS) March 2024 - March 2025
 - Development of industrial defect segmentation model that captures small defects in production.

OTHER PUBLICATIONS

- **Tunable Porous Collagen Hydrogels as a Physiologically Relevant Platform for Extrachromosomal DNA-Associated Colorectal Cancer Research**
S Jo, J Shon, S An, Y Nam, D Choi, S Lee, **Hoigi Seo**, S Y Chun & H Kim.
Theranostics (IF: 13.3), 2025.
- **ecSegCls: Deep learning-based method for detecting extrachromosomal DNAs in both interphase and metaphase cancer cells**
Hoigi Seo, Y Nam, H Bae, D U Kang, J Lee, H Kim & S Y Chun.
American Association for Cancer Research (**AACR**), 2026.
- **Harmonization for a black-box deep learning model**
M Kim, H Jeong, **Hoigi Seo**, W Jeong, S Y Chun & J Lee.
International Society for Magnetic Resonance in Medicine (**ISMRM**) (**Oral, Summa Cum Laude, AMPC select**), 2025
- **Mitigating Hallucinations in Large Vision-Language Models via Norm-Guided Visual Filtering**
H Cho, **Hoigi Seo**, D U Kang, J Lee & S Y Chun.
Conference on Electronics, Semiconductor and AI, 2025.
- **Towards Personalization of Generative Models Via Zeroth-Order Optimization**
W Jeong, **Hoigi Seo** & S Y Chun.
Korea Signal Processing Conference, 2024.

ACADEMIC SERVICE

- **CVPR, ECCV, NeurIPS, WACV** Reviewer
• *Served as a reviewer for top-tier AI conferences.* 2026 - Present

MEDIA

- **Erasing 2,072 concepts on pre-trained diffusion model** Interviewee
• *Popular Science Korea* June 2026
 - Discussed ETC, our CVPR 2026 method that erases 2,072 concepts (949 celebrities, 693 art styles, 430 characters) from a pre-trained diffusion model without manual tuning, and its implications for safety- and copyright-driven AI control.

INVITED TALK

- **On Efficient Diffusion Model Workflow** Invited Speaker
• *Samsung MX Seminar* April 2026
 - Presented a training-free technique that generates low-resolution previews perceptually consistent with the final high-resolution image, enabling early candidate selection and up to 3× faster diffusion model workflows.

OTHER EXPERIENCE

- **Military Service** Sergeant
• *Served as a Situation Room Soldier in the 3rd Guard Battalion.* May 2020 - November 2021